

Problem

Evaluate the following complex numbers:

$$(a) \frac{2+j3}{1-j6} + \frac{7-j8}{-5+j11}$$

$$(b) \frac{(5\angle 10^\circ)(10\angle -40^\circ)}{(4\angle -80^\circ)(-6\angle 50^\circ)}$$

$$(c) \begin{vmatrix} 2+j3 & -j2 \\ -j2 & 8-j5 \end{vmatrix}$$

Step-by-step solution

Step 1 of 3

$$\begin{aligned} (A) \quad & \frac{2+j3}{1-j6} + \frac{7-j8}{-5+j11} \\ &= \frac{(2+j3)(1+j6)}{37} + \frac{(7-j8)(-5-j11)}{146} \\ &= \frac{(2-18)+j(12+3)}{37} + \frac{(-35-88)+j(-77+40)}{146} \\ &= -0.432 + j0.4054 - 0.842 - j0.253 \\ &= -1.274 + j0.1524 \end{aligned}$$

Step 2 of 3

$$\begin{aligned} (B) \quad & \frac{(5\angle 10^\circ)(10\angle -40^\circ)}{(4\angle -80^\circ)(-6\angle 50^\circ)} \\ &= \frac{50\angle -30^\circ}{-24\angle -30^\circ} \\ &= -2.083\angle -30^\circ + 30^\circ \\ &= -2.083\angle 0^\circ \end{aligned}$$

Step 3 of 3

$$\begin{aligned} (C) \quad & \begin{vmatrix} 2+j3 & -j2 \\ -j2 & 8-j5 \end{vmatrix} \\ &= (2+j3)(8-j5) - (-j2)(-j2) \\ &= (16+15) + j(-10+24) + 4 \\ &= 35 + j14 \end{aligned}$$